Reporting on the Cathedral: A Guide for Science Journalists

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Abstract

This document provides accurate framing for journalists covering the Cathedral project: a machine-verified reduction of the Riemann Hypothesis in Lean 4. It contains: (1) what the project is, in one paragraph; (2) what it is *not*, which is more important; (3) a fact sheet with verified numbers; (4) common misconceptions and their corrections; (5) suggested expert sources for independent evaluation; and (6) quotes approved for use. The primary goal of this document is to prevent misreporting. The Riemann Hypothesis has **not been proved**. Cryptography has **not been broken**. What has been built is significant in its own right, and accurate reporting will serve the public better than sensationalism.

1 The One-Paragraph Summary

A computer scientist in New Mexico, working with AI assistance over 23 days, built a 78-file computer program in the Lean 4 theorem prover that establishes, with machine-verified certainty, that the 167-year-old Riemann Hypothesis is logically equivalent to the convergence of a concrete mathematical approximation problem. The proof is complete in both directions and relies on exactly two mathematical assumptions ("axioms"), both of which are well- understood results in classical mathematics that have not yet been translated into computer-checkable form. The project does not prove the Riemann Hypothesis. It provides the most precise formal map ever made of what a proof would require.

2 What It Is NOT

This section is more important than the previous one.

1. **NOT a proof of the Riemann Hypothesis.** The project reduces RH to two axioms. It does not prove those axioms. The distinction is critical: a reduction is a map of the problem, not a solution to it.

2. **NOT a threat to cryptography.** A companion paper (*cathedral-security.tex*) establishes rigorously that the mathematical tools used in the Cathedral provide no computational advantage for breaking encryption. Even a full proof of RH would not break RSA, AES, or post-quantum cryptography. Headlines like "Riemann Hypothesis breakthrough threatens internet security" would be factually wrong.

3. **NOT AI solving mathematics autonomously.** The AI generated code under human direction. The human provided architectural vision, quality control, and mathematical taste. The AI did not "solve" anything—it implemented strategies designed by the human and verified by the Lean compiler.

4. **NOT a claim to the Millennium Prize.** The Clay Mathematics Institute requires a proof of RH itself, not an equivalence. This project would not qualify.

5. **NOT a professional mathematics paper.** The author is a computer scientist, not a professional mathematician. The project is a formalization (software engineering applied to mathematics), not a traditional research publication.

3 What It IS

1. **A machine-verified formal proof.** Every logical step has been checked by the Lean 4 compiler. There are zero errors, zero gaps, and zero unchecked assumptions beyond the two stated axioms.

2. **A complete biconditional.** The equivalence goes in both directions: RH implies convergence (forward), and convergence implies RH (converse). Both are verified.

3. **A case study in human-AI collaboration.** One person with AI assistance produced in 23 days what would traditionally require a research team over months. This has implications for how mathematical research is conducted.

4. **A transparent artifact.** All 78 active files, 96 archived files, 40 axioms, and 644 theorems are available for inspection. Everything is documented.

4 Fact Sheet

Item	Verified Value
Active Lean files	78
Archived Lean files	96
Total axioms (active)	40
Crown axioms (on critical path)	2
Machine-verified theorems	644
Compilation errors	0
Development time	23 days
Team size	1 human + AI assistants
Prover	Lean 4
Mathematical library	Mathlib
Companion papers	21
Millennium Prize eligibility	No
Cryptographic impact	None

5 Common Misconceptions — Corrected

Wrong	Right
"The Riemann Hypothesis has been proved."	The RH has been *reduced* to two well-understood axioms. It has not been proved.
"This breaks encryption."	A companion security paper proves it does not. Even a full proof of RH would not break modern cryptography.
"AI solved a famous math problem."	A human used AI as a coding assistant to build a formal verification system. The Lean compiler verified it.
"This is just a computer program, not real math."	Machine-verified proofs are *more* rigorous than traditional proofs: every step is checked, zero gaps.
"The project uses 40 unproved assumptions."	40 axioms total, but only 2 are on the crown theorem's critical path. The other 38 support alternative proof routes.
"This was done in 23 days, so it can't be serious."	The tools (Lean 4, Mathlib, AI) have transformed what is possible. The result is 644 verified theorems, not a hasty sketch.

6 The Riemann Hypothesis — Explained Simply

For readers unfamiliar with the mathematics:

Prime numbers (2, 3, 5, 7, 11, 13, ...) are the building blocks of all whole numbers. Every number can be written as a product of primes. We know roughly how many primes there are below any given number, but we don't know their exact pattern.

In 1859, Bernhard Riemann found a mathematical function (the "zeta function") that encodes where the primes are. This function has special values called "zeros" that control the pattern of primes. Riemann conjectured that all these zeros line up perfectly on a single line in the complex plane.

If true, primes are as orderly as we hope. If false, there is deep irregularity in the fabric of arithmetic.

The Cathedral proves that this conjecture is *exactly equivalent* to whether certain wiggly curves (sawtooth waves) can be combined to make a flat line. Both directions of this equivalence are machine-verified.

7 Suggested Expert Sources

For independent evaluation, journalists may wish to consult:

- **Lean/Mathlib community**: The Lean Zulip chat (`https://leanprover.zulipchat.com/`) includes experts who can evaluate the formal proof.

- **Analytic number theory**: Any university faculty member specializing in analytic number theory can evaluate whether the two axioms are standard results.
- **Formal verification**: Experts at Carnegie Mellon, Imperial College, or Microsoft Research can evaluate the verification methodology.
- **AI and mathematics**: Google DeepMind (AlphaProof team) or equivalent groups can evaluate the human-AI collaboration claims.

8 Approved Quotes

The following quotes may be attributed to the author:

> "We did not prove the Riemann Hypothesis. We built a machine- verified map of exactly what a proof would require. The map has two blank spots. Those blank spots are the hardest parts."
>
> "The Lean compiler doesn't care who you are. It only cares whether your proof is correct. That's what makes machine verification powerful: it's the most objective form of mathematical truth we have."
>
> "This project does not break cryptography, does not solve a Millennium Prize Problem, and does not mean AI is replacing mathematicians. What it does mean is that one person with the right tools can contribute to frontier mathematics. That's new."

9 What NOT to Write

Please avoid:

- Headlines containing "proved," "solved," or "cracked" in reference to the Riemann Hypothesis.
- Any implication that encryption or internet security is affected.
- Framing the AI as an independent agent that "discovered" mathematical results.
- Comparisons to the Millennium Prize that imply eligibility.
- Use of the word "breakthrough" without the qualification "in formalization methodology."

Accurate alternatives:

- "Machine-verified reduction of the Riemann Hypothesis"
- "Formal proof maps famous conjecture to two axioms"
- "Human-AI collaboration produces verified math in 23 days"
- "Computer checks every step of 167-year-old problem's logical structure"

10 Contact

For questions, clarifications, or interview requests, contact the author directly. All companion papers are available upon request.